MRScoreg

User manual and documentation

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Developed by: Lili Fanni Tóth at MIDAlab, Institute of Biotechnology and Biomedicine (IBB), Universitat Autònoma de Barcelona (UAB)

Guthub: https://github.com/lilively/mrsplotter

Overview:

MRScoreg is a user-friendly software tool designed to facilitate the integration of magnetic resonance spectroscopy (MRS) data with magnetic resonance imaging (MRI). Spatial parameters are extracted from DICOM headers, and a sequence of processing steps generates anatomical image regions corresponding to each voxel.

The software is open source, with its source code freely accessible on GitHub under the CC BY-NC-SA 4.0 Plus license (https://github.com/lilively/mrscoreg). A compiled executable version is available for users who prefer direct installation or don't have a MATLAB license. The application includes precompiled SPM12 functions and automatically configures ITK-SNAP C3D on first launch. For users requiring access to the source code, the complete MATLAB application (.mlapp) is available on GitHub (https://github.com/lilively/mrscoreg). This allows users with MATLAB R2025a or later to modify and customize.

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Introduction

A key step of magnetic resonance spectroscopy (MRS) analysis is the accurate extraction and characterization of anatomical regions corresponding to spectroscopy voxels. MRSHub[1] provides several options for MRS data visualization. For instance, SIVIC[2] offers robust MRSI spatial visualization capabilities, but its primary emphasis is on spectral analysis and metabolite mapping rather than on extracting and averaging anatomical content from individual voxel regions. In contrast, both Osprey[3] and Gannet[4] include segmentation features; however, their primary focus is on tissue fraction correction for quantification, rather than detailed spatial extraction and averaging of voxel anatomical content.

MRScoreg is a software tool that helps integrate magnetic resonance spectroscopy (MRS) data with magnetic resonance imaging (MRI). The application automatically extracts the acquisition parameters from the DICOM headers and uses the voxel dimensions and positioning information to compute the spectroscopy voxel coordinates. The main interface(

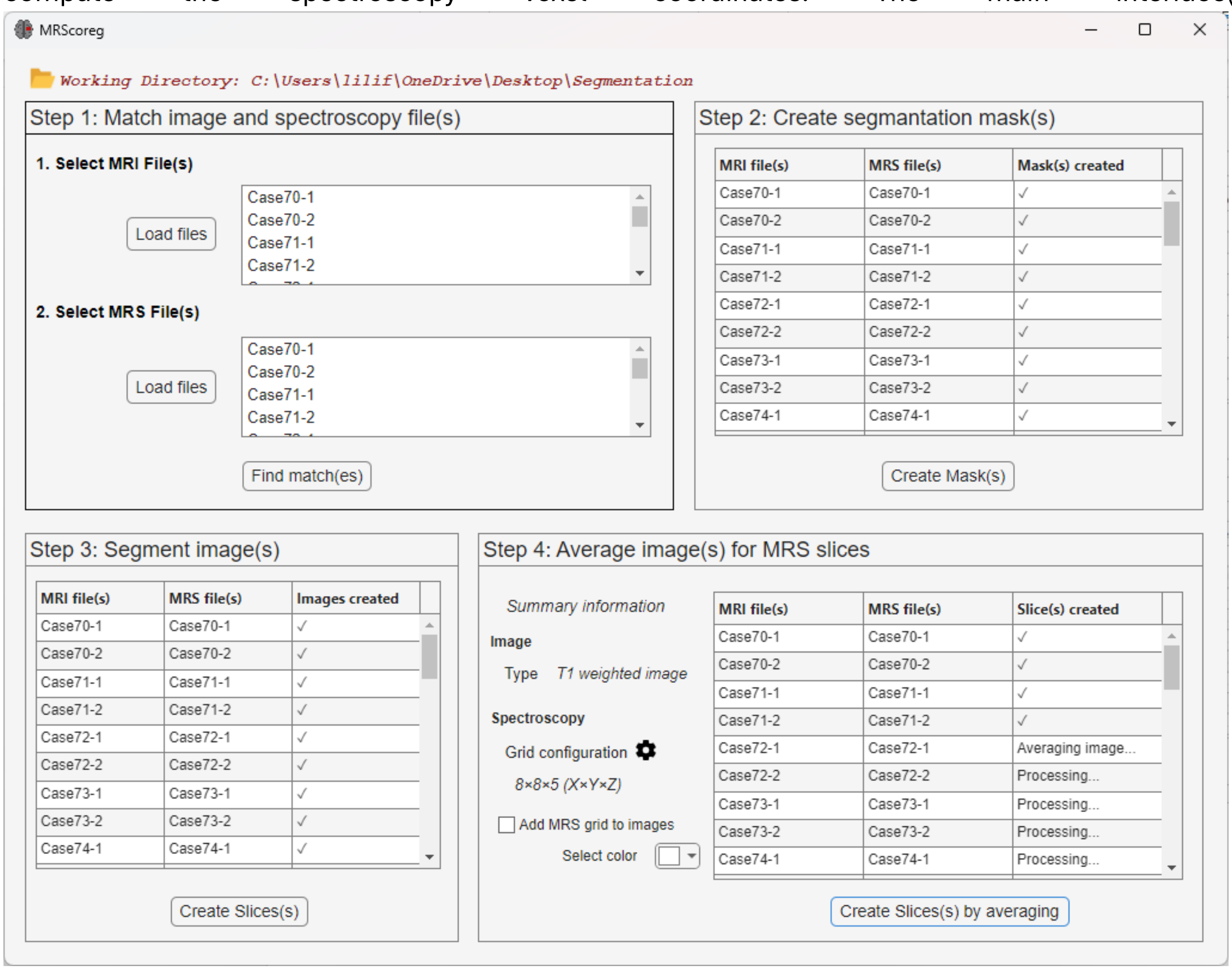

Figure 1) is divided into four panels indicating the processing steps. The output directory can be changed via the folder button located above the upper panels. The accompanying console provides real-time information about the operations executed.

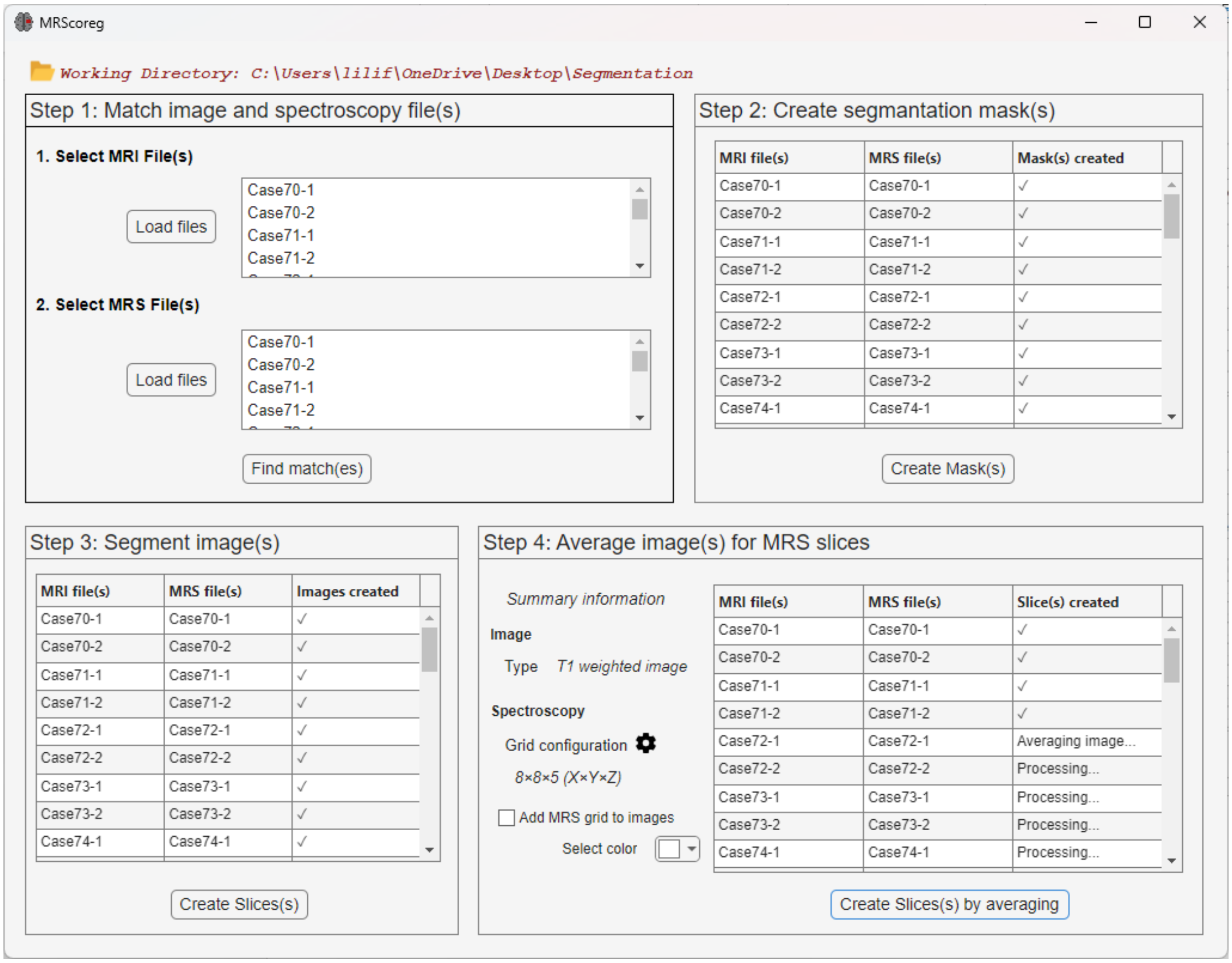

Figure 1 Main interface.

Requirements

- Operating System: Windows 10 or 11 (may work on Windows 7/8)
- RAM: Minimum 4GB recommended (8GB+ for larger datasets)
- Disk Space: At least 200MB free space
- Permissions: Write access to folders where you intend to save results

Supported file formats

Digital Imaging and Communications in Medicine (DICOM)

- T1 weighted images
- Spectroscopy files

Installation and setup

Windows version (Standalone Executable)

After downloading the latest version from the GitHub releases page, extract the contents and start the application by double-clicking on the MRScoreg.exe. The MATLAB Runtime 2025a will be automatically retrieved during the installation process if it is not detected in the system. This component is available free of charge and does not require a MATLAB license.

The program uses Convert3D, a command-line image conversion and processing tool version of ITKSNAP[6,7,9]. Upon the first execution, a configuration file (Config.txt) is generated (**Error! Reference source not found.**). This needs to be modified with the Convert3 installed path (**Error! Reference source not found.**). Subsequent executions will be configured automatically (**Error! Reference source not found.**).

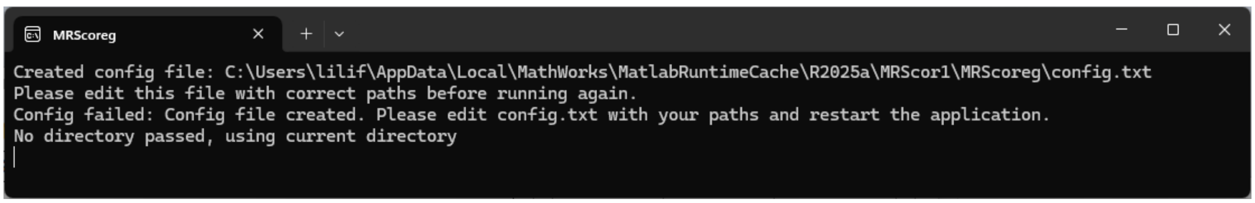

Figure 2 Upon the first run, a configuration file is generated.

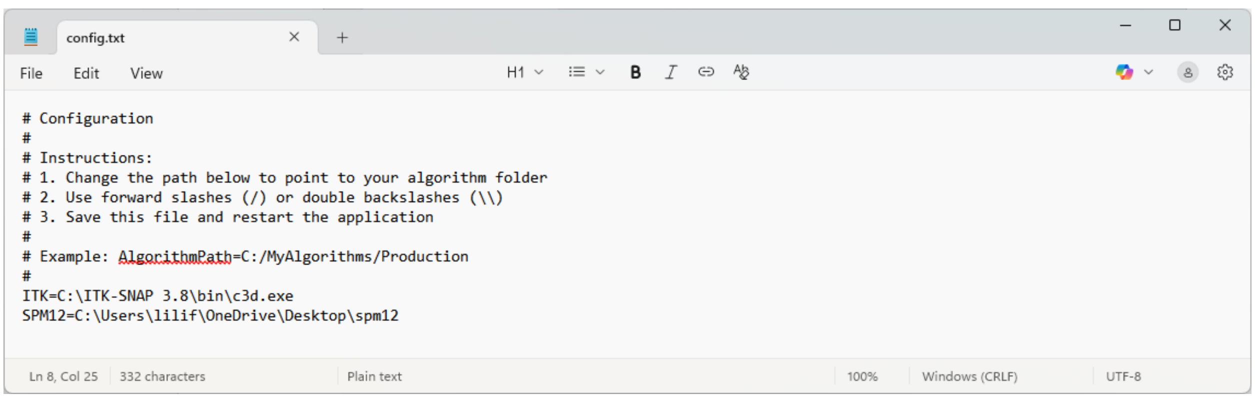

Figure 3 Generated configuration text file.

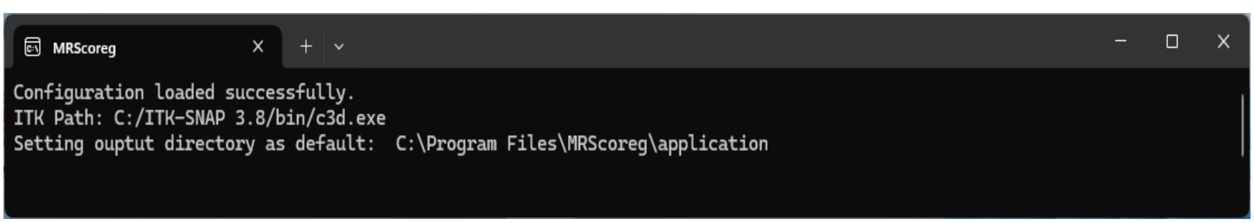

Figure 4 From the next run, the configuration is automatic.

Source Code Distribution

Complete source code can be found on GitHub. To get it, clone the repository to your local machine:

```
git clone https://github.com/lilively/mrscoreg
```

To work with the source code, open the MRScoreg.mlapp file using MATLAB App Designer in MATLAB R2025a or later. The application can then be executed or modified within the MATLAB environment. On first launch, follow the configuration steps described in section 0.

Step 1: Match image and spectroscopy File(s)

First the image and spectroscopy files need to be loaded via the "Load files" buttons. By clicking on the "Find match(es)" button the image and spectroscopy files are matched up during the unique Study Instance UID Attribute [10] parameter read from the DICOM files. After matching the image and spectroscopy files, the necessary acquisition parameters are extracted (Table 1).

Table 1 Dicom tags used for localization.

Private Tag	Tag	Desciption
(2005, 1074)	MRStackFovAP	Field of view anterior-posterior
(2005, 1076)	MRStackFovRL	Field of view right-left
(2005, 1075)	MRStackFovFH	Field of view food-head
(2005, 1078)	MRStackOffcentreAP	Offset anterior-posterior
(2005, 107A)	MRStackOffcentreRL	Offset right-left
(2005, 1079)	MRStackOffcentreFH	Offset foot-head
(2005, 1071)	MRStackAngulationAP	Angulation anterior-posterior
(2005, 1073)	MRStackAngulationRL	Angulation right-left
(2005, 1072)	MRStackAngulationFH	Angulation foot-head

Step 2: Create a binary mask

First the NIfTI The DICOM MRI image is then converted to NIfTI format using Covert3D[6], a command-line implementation of ITK-SNAP[7]. First delaunay triangulation is performed using SPM12[8] to achieve precise alignment between MRS and MRI acquisitions. From the alignment, a binary mask is generated in NIfTI format, which can be directly opened in ITK-SNAP [7] or medical image processing software(Figure 5). The generated files are shown on Figure 6.

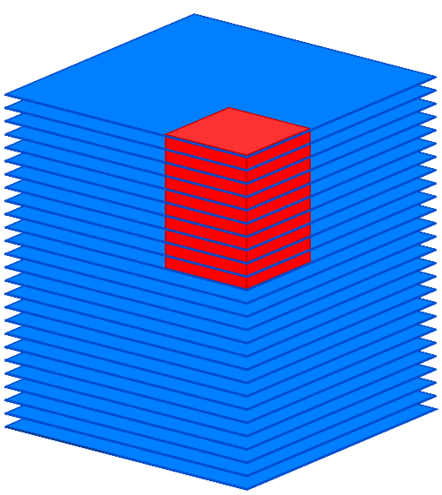

Figure 5 Representation of the binary mask.

```
Segmentation_output
├── NIFTI_Files/
│   ├── Case_1
│   ├── Case_2
│   └── Case_n
└── Masks/
    ├── Case_1
    ├── Case_2
    └── Case_n
```

Figure 6 Files produced from step 2.

Step 3: Segment image(s)

The mask generated in the previous step is applied to the MRI image, enabling the extraction of the masked region as a stack of photosFigure 7. The generated files are shown on Figure 8.

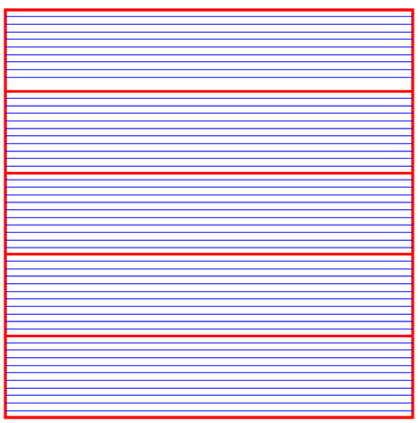

Figure 7 Representation of a stack of MRI images corresponding to 5 slices or MRSI acquisitions.

```
Segmentation_output/
├── NIFTI_Files/
│   ├── Case_1
│   ├── Case_2
│   └── Case_3
├── Masks/
│   ├── Case_1
│   ├── Case_2
│   └── Case_3
└── Segmented images/
    ├── Case_1
    │   └ All
    ├── Case_2
    │   └ All
    └── Case_3
        └ All
```

Figure 8 Files produced from step 3.

Step 4: Average image(s)

The stack of images can be averaged to produce a representative composite image, while individual images remain accessible (Figure 9).

Optionally, the spectroscopy voxel grid may be overlaid on the segmented, averaged image to facilitate interpretation of spectral data relative to anatomical structures. The row and column values can be modified if the automatic acquisition fails. The gear icon can be used to do this. The grid color can also be customized by selecting a color. Figure 10 shows two generated images.

```
Segmentation_output/
├── NIFTI_Files/
│   ├── Case_1
│   ├── Case_2
│   └── Case_3
├── Masks/
│   ├── Case_1
│   ├── Case_2
│   └── Case_3
└── Segmented images/
    ├── Case_1
    │   ├ All
    │   ├ Slice 1
    │   ├ Slice 2
    │   └ Slice n
    ├── Case_2
    │   ├ All
    │   ├ Slice 1
    │   ├ Slice 2
    │   └ Slice n
    └── Case_3
        ├ All
        ├ Slice 1
        ├ Slice 2
        └ Slice n
```

Figure 9 Files produced from step 4.

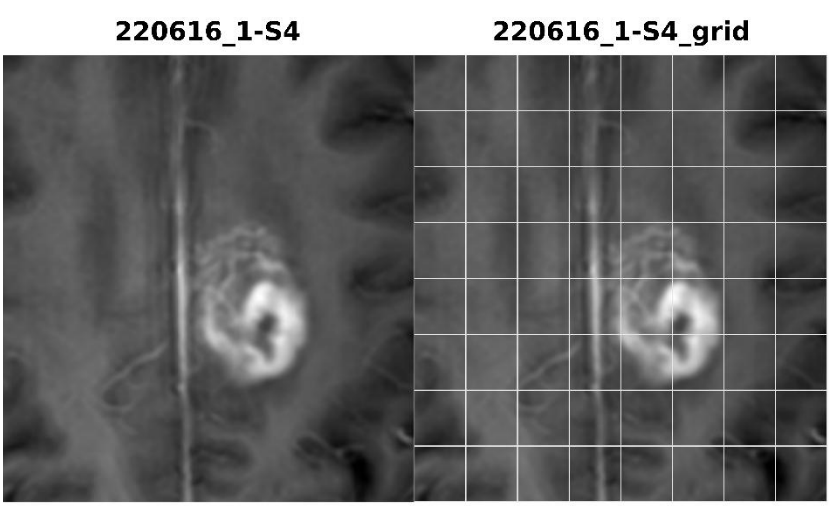

Figure 10 Example of anatomical images generated. (Left): no gird applied, (Right): automatic grid configuration applied.

References

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